



NEET(UG)
MAJOR

This Booklet contains 24 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet, fill in the particulars on Side-1 and Side-2 carefully with blue/black ball point pen only.
2. The test is of 3 hours duration and this Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
3. Use Blue/Black Ball Point Pen only for writing particulars on this page/marking responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
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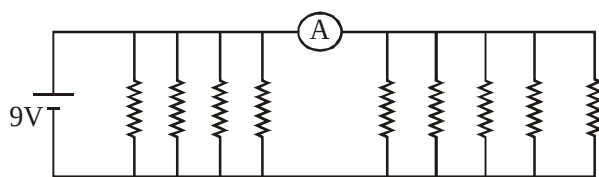
Facsimile signature stamp of

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LEADER & ACHIEVER COURSE
(PHASE : MLI,J,K,M,N,O,R,S,M4L8A,MAZG,H,I,J,K,L,M,T,U,M1A5A,
M2A5B,M4AA2A,M2AP1A,M2AP1B)

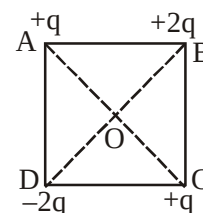
ALLEN MAJOR TEST		DATE : 19 - 03 - 2020
SYLLABUS – 03 & 04		
PHYSICS	:	Thermal Physics (Thermal Expansion, Calorimetry, Heat Transfer, KTG & Thermodynamics), Oscillations (SHM, Damped and Forced Oscillations & Resonance), Wave Motion and Doppler's Effect., Electrostatics, Capacitor, Current electricity and Heating Effects of Current.
CHEMISTRY	:	Classification & Nomenclature, Isomerism, Reaction Mechanism-I, Reaction Mechanism-II, Hydrocarbons, Environmental Chemistry and Qualitative and Quantitative Analysis of Organic Compounds, Solid State, Chemical kinetics, Solutions, Electrochemistry, Surface chemistry
BIOLOGY	:	Human Physiology : (i) Digestion and Absorption (ii) Breathing and Exchange of Gases (iii) Body Fluids and circulation (iv) Excretory Products and their Elimination (v) Locomotion and Movement (vi) Neural Control and Coordination, Eye & Ear (vii) Chemical Coordination and Integration (viii) Animal Kingdom, Cockroach, Earthworm, Frog (ix) Structural Organisation in Animal (Animal tissue), Reproduction : (i) Reproduction in Organisms (ii) Sexual Reproduction in Flowering Plants (iii) Human Reproduction (iv) Reproductive Health (v) Evolution

1. A steel ball of mass 0.1 kg falls freely from a height of 10 m and bounces to a height of 5.4 m from the ground. If the dissipated energy in this process is absorbed by the ball, the rise in its temperature is :-
(Specific heat of steel = $460 \text{ Joule} \cdot \text{kg}^{-1} \cdot ^\circ\text{C}^{-1}$, $g = 10 \text{ ms}^{-2}$)
(1) 0.01°C
(2) 0.1°C
(3) 1°C
(4) 1.1°C
2. A sample of gas is at 0°C . To what temperature it must be raised in order to double the r.m.s. speed of the molecule :-
(1) 270°C (2) 819°C (3) 1090°C (4) 100°C
3. A man sets his watch by the sound of a siren placed at a distance 1 km away. If the velocity of sound is 330 m/s :-
(1) His watch is set 3 sec. faster
(2) His watch is set 3 sec. slower
(3) His watch is set correctly
(4) None of the above
4. Two equal -ve charges $-q$ are fixed at the points (0, a) and (0, -a) on the y-axis. A positive charge Q is released from rest at the point (2a, 0) on the x-axis. The charge will :-
(1) execute SHM about the origin
(2) move to the origin and remain at rest
(3) move to infinity
(4) execute oscillatory but not SHM
5. If each resistance in the figure of 9Ω then reading of the ammeter is :-



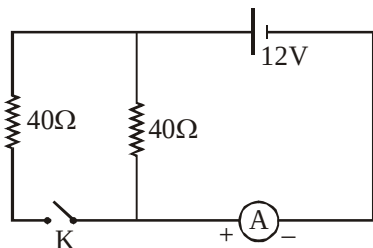
- (1) 8A (2) 5A
(3) 2A (4) 9A

6. A system performs work ΔW when an amount of heat is ΔQ added to the system, the corresponding change in the internal energy is ΔU . A unique function of the initial and final states (irrespective of the mode of change) is :-
(1) ΔQ (2) ΔW
(3) ΔU and ΔQ (4) ΔU
7. For a gas if $\gamma = 1.4$, then atomicity, C_p and C_v of the gas are respectively :-
(1) Monoatomic, $\frac{5}{2}R, \frac{3}{2}R$
(2) Monoatomic, $\frac{7}{2}R, \frac{5}{2}R$
(3) Diatomic, $\frac{7}{2}R, \frac{5}{2}R$
(4) Triatomic, $\frac{7}{2}R, \frac{5}{2}R$
8. Speed of sound in mercury at a certain temperature is 1450 m/s . Given the density of mercury as $13.6 \times 10^3 \text{ kg/m}^3$, the bulk modulus for mercury is :-
(1) $2.86 \times 10^{10} \text{ N/m}^2$
(2) $3.86 \times 10^{10} \text{ N/m}^2$
(3) $4.86 \times 10^{10} \text{ N/m}^2$
(4) $5.86 \times 10^{10} \text{ N/m}^2$
9. Four charges are arranged at the corners of a square ABCD as shown in the figure. The force on the charge kept at the centre O is :-



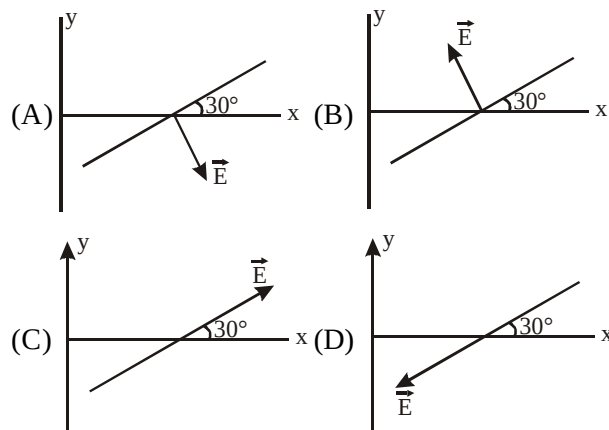
- (1) zero
(2) along the diagonal AC
(3) along the diagonal BD
(4) perpendicular to side AB

10. In the given circuit emf of cell = 12V & internal resistance of cell is zero. Reading of A is i_1 when key is opened & reading of A is i_2 when key is closed then $\frac{i_1}{i_2}$ will be :-



- (1) 2 (2) 0.3
(3) 0.6 (4) 0.5
11. In an isothermal reversible expansion, if the volume of 96 gm of oxygen at 27°C is increased from 70 litres to 140 litres, then the work done by the gas will be :-
(1) $300 R \log_{10} 2$
(2) $81 R \log_e 2$
(3) $900 R \log_{10} 2$
(4) $2.3 \times 900 R \log_{10} 2$
12. If the intermolecular forces vanish away, the volume occupied by the molecules contained in 4.5 kg water at standard temperature and pressure will be :-
(1) 5.6 m^3 (2) 4.5 m^3
(3) 11.2 litre (4) 11.2 m^3
13. A tuning fork whose frequency as given by manufacturer is 512 Hz is being tested with an accurate oscillator. It is found that the fork produces a beat of 2 Hz when oscillator reads 514 Hz but produces a beat of 6 Hz when oscillator reads 510 Hz. The actual frequency of fork is :-
(1) 508 Hz
(2) 512 Hz
(3) 516 Hz
(4) 518 Hz

14. Some equipotential surfaces, which are normal to x-y plane are shown in the adjoining figure the direction of the electric field is:



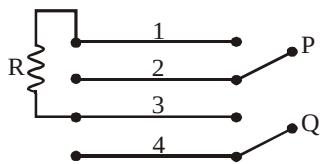
Options :-

- (1) A or B (2) A or C
(3) C or D (4) B or D
15. For a cell of emf 2V, a balance is obtained for 50 cm of the potentiometer wire. If the cell is shunted by a 2Ω resistor, the balance is obtained across 40 cm of the wire. Find the internal resistance of the cell :-
(1) 0.25Ω (2) 0.50Ω (3) 0.80Ω (4) 1.00Ω
16. A monoatomic gas ($\gamma = 5/3$) is suddenly compressed to $\frac{1}{8}$ of its original volume adiabatically, then the pressure of the gas will change to :-
(1) $\frac{24}{5}$ times its initial pressure
(2) 8 times its initial pressure
(3) $\frac{40}{3}$ times its initial pressure
(4) 32 times its initial pressure
17. Two identical plates of different metals are joined to form a single plate whose thickness is double the thickness of each plate. If the coefficients of conductivity of each plate are 2 and 3 respectively, then the conductivity of composite plate will be :-
(1) 5 (2) 2.4
(3) 1.5 (4) 1.2

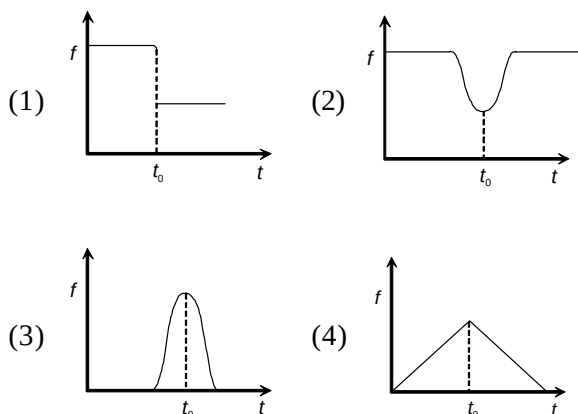
18. Four wires of identical length, diameters and of the same material are stretched on a sonometre wire. If the ratio of their tensions is 1 : 4 : 9 : 16 then the ratio of their fundamental frequencies are :-
 (1) 16 : 9 : 4 : 1 (2) 4 : 3 : 2 : 1
 (3) 1 : 4 : 2 : 16 (4) 1 : 2 : 3 : 4
19. If the electric field is given by $(5\hat{i} + 4\hat{j} + 9\hat{k})$, the electric flux through a surface of area 20 unit lying in the Y-Z plane will be:
 (1) 100 unit (2) 80 unit
 (3) 180 unit (4) 20 unit
20. A galvanometer of resistance G is shunted by a resistance S ohm. To keep the main current in the circuit unchanged the resistance to be put in series with the galvanometer is :-
 (1) $\frac{G^2}{S+G}$ (2) $\frac{G}{S+G}$ (3) $\frac{S^2}{S+G}$ (4) $\frac{SG}{S+G}$
21. One mole of a perfect gas in a cylinder fitted with a piston has a pressure P, volume V and temperature T. If the temperature is increased by 1 K keeping pressure constant, the increase in volume is :-
 (1) $\frac{2V}{T}$ (2) $\frac{3V}{T}$ (3) $\frac{V}{T}$ (4) V
22. A body radiates energy 5W at a temperature of 127°C. If the temperature is increased to 927°C, then it radiates energy at the rate of :-
 (1) 410 W (2) 81 W (3) 405 W (4) 200 W
23. Calculate the frequency of the second harmonic formed on a string of length 0.5 m and mass 2×10^{-4} kg when stretched with a tension of 20 N :-
 (1) 274.4 Hz (2) 744.2 Hz
 (3) 44.72 Hz (4) 447.2 Hz
24. The electric potential at a point (x, 0, 0) is given by:

$$V = \left[\frac{1000}{x} + \frac{1500}{x^2} + \frac{500}{x^3} \right]$$
 then the electric field at x = 1 m is (in volt/m) :
 (1) $-5500\hat{i}$ (2) $5500\hat{i}$
 (3) $\sqrt{5500}\hat{i}$ (4) zero
25. A parallel plate capacitor has the space between its plates filled by the two slabs of thickness $\frac{d}{2}$ each and dielectric constants K_1 and K_2 , d is the plate separation of capacitors. The capacity of the capacitor is :-
 (1) $\frac{2\epsilon_0 d}{A} \left(\frac{K_1 + K_2}{K_1 K_2} \right)$ (2) $\frac{2\epsilon_0 A}{d} \left(\frac{K_1 K_2}{K_1 + K_2} \right)$
 (3) $\frac{2\epsilon_0 A}{d} (K_1 + K_2)$ (4) $\frac{2\epsilon_0 A}{d} \left(\frac{K_1 + K_2}{K_1 K_2} \right)$
26. A Carnot engine whose low temperature reservoir is at 7°C has an efficiency of 50%. It is desired to increase the efficiency to 70%. By how many degrees should the temperature of the high temperature reservoir be increased :-
 (1) 840 K (2) 280 K (3) 560 K (4) 380 K
27. If the displacement of a particle executing SHM is given by $y = 0.30 \sin(220t + 0.64)$ in metre, then the frequency and maximum velocity of the particle is :-
 (1) 35 Hz, 66 m/s (2) 45 Hz, 66 m/s
 (3) 58 Hz, 113 m/s (4) 35 Hz, 132 m/s
28. What is minimum length of a tube, open at both ends, that resonates with tuning fork of frequency 350 Hz ? [velocity of sound in air = 350 m/s] :-
 (1) 50 cm
 (2) 100 cm
 (3) 75 cm
 (4) 25 cm
29. There concentric metallic spherical shells of radii R, 2R, 3R are given charges Q_1 , Q_2 , Q_3 , respectively. It is found that the surface charge densities on the outer surface of the shells are equal. Then, the ratio of the charges given to the shells, Q_1 , Q_2 , Q_3 , is :
 (1) 1 : 2 : 3
 (2) 1 : 3 : 5
 (3) 1 : 4 : 9
 (4) 1 : 8 : 18

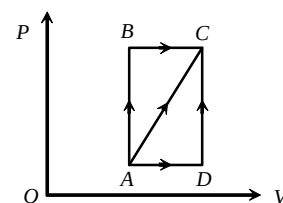
30. Four metallic plates, each with surface area of one side A, are placed at a distance d from each other. The plates are connected as shown in the adjoining figure. Then the capacitance of the system between P and Q is :-



- (1) $\frac{3\epsilon_0 A}{d}$ (2) $\frac{2\epsilon_0 A}{d}$
 (3) $\frac{2\epsilon_0 A}{3d}$ (4) $\frac{3\epsilon_0 A}{2d}$
31. The coefficient of performance of a Carnot refrigerator working between 30°C and 0°C is :-
 (1) 10 (2) 1 (3) 9 (4) 0
32. The P.E. of a particle executing SHM at a distance x from its equilibrium position is
 (1) $\frac{1}{2} m\omega^2 x^2$ (2) $\frac{1}{2} m\omega^2 a^2$
 (3) $\frac{1}{2} m\omega^2 (a^2 - x^2)$ (4) Zero
33. A man is standing on a railway platform listening to the whistle of an engine that passes the man at constant speed without stopping. If the engine passes the man at time t_0 . How does the frequency f of the whistle as heard by the man changes with time :-



34. A current of 5 amp exists in a 10 ohm resistance for 4 min. How many coulombs pass through any cross-section of the resistor in this time ?
 (1) 12 (2) 120 (3) 1200 (4) 12000
35. Four capacitors of equal capacitance have an equivalent capacitance C_1 when connected in series and an equivalent capacitance C_2 when connected in parallel. The ratio $\frac{C_1}{C_2}$ is :-
 (1) $\frac{1}{4}$ (2) $\frac{1}{16}$
 (3) $\frac{1}{8}$ (4) $\frac{1}{12}$
36. A thermodynamic process is shown in the figure. The pressures and volumes corresponding to some points in the figure are :
 $P_A = 3 \times 10^4 \text{ Pa}$, $P_B = 8 \times 10^4 \text{ Pa}$ and
 $V_A = 2 \times 10^{-3} \text{ m}^3$, $V_D = 5 \times 10^{-3} \text{ m}^3$
 In process AB, 600 J of heat is added to the system and in process BC, 200 J of heat is added to the system. The change in internal energy of the system in process AC would be :-

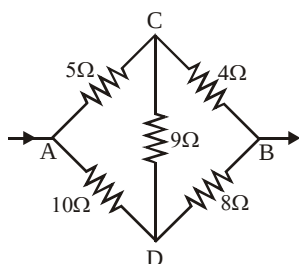


- (1) 560 J (2) 800 J (3) 600 J (4) 640 J
37. The period of simple pendulum is measured as T in a stationary lift. If the lift moves upwards with an acceleration of 5 g, the period will be :-
 (1) The same
 (2) Increased by $\frac{3}{5}$
 (3) Decreased by $\frac{2}{3}$ times
 (4) None of the above

38. A $10\ \mu\text{F}$ capacitor and a $20\ \mu\text{F}$ capacitor are connected in series across a $200\ \text{V}$ supply line. The charged capacitors are then disconnected from the line and reconnected with their positive plates together and negative plates together and no external voltage is applied. What is the potential difference across each capacitor

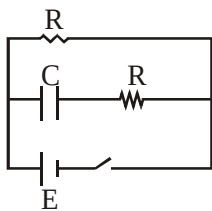
- (1) $\frac{400}{9}\ \text{V}$ (2) $\frac{800}{9}\ \text{V}$
(3) $400\ \text{V}$ (4) $200\ \text{V}$

39. Five resistors are connected as shown in the diagram. The equivalent resistance between A and B is



- (1) $6\ \text{ohm}$
(2) $9\ \text{ohm}$
(3) $12\ \text{ohm}$
(4) $15\ \text{ohm}$

40. In given figure at $t = 0$ key 's' is closed then find charge on capacitor when current through cell is 50% of maximum current :-

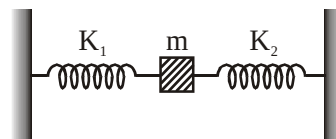


- (1) $\frac{EC}{3}$ (2) $\frac{EC}{2}$
(3) $\frac{EC}{4}$ (4) EC

41. Three containers of the same volume contain three different gases. The masses of the molecules are m_1 , m_2 and m_3 and the number of molecules in their respective containers are N_1 , N_2 and N_3 . The gas pressure in the containers are P_1 , P_2 and P_3 respectively. All the gases are now mixed and put in one of the containers. The pressure P of mixture will be :-

- (1) $P < (P_1 + P_2 + P_3)$ (2) $P = \frac{P_1 + P_2 + P_3}{3}$
(3) $P = P_1 + P_2 + P_3$ (4) $P > (P_1 + P_2 + P_3)$

42. The effective spring constant of two spring system as shown in figure will be :-



- (1) $K_1 + K_2$ (2) $K_1 K_2 / (K_1 + K_2)$
(3) $K_1 - K_2$ (4) $K_1 K_2 / (K_1 - K_2)$

43. If a body gives out $10^9\ e^-$ per second, how much time is required to get a total charge of $1\ \text{C}$ from it?

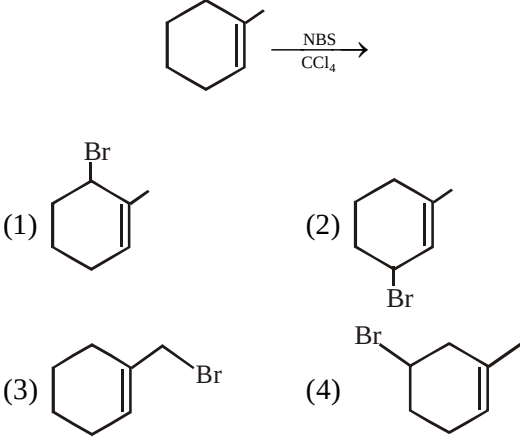
- (1) Around 198 minute
(2) Around 198 h
(3) Around 198 days
(4) Around 198 year

44. Four resistances $10\ \Omega$, $5\ \Omega$, $7\ \Omega$ and $3\ \Omega$ are connected so that they form the sides of a rectangle AB, BC, CD and DA respectively. Another resistance of $10\ \Omega$ is connected across the diagonal AC. The equivalent resistance between A and B is :-

- (1) $2\ \Omega$ (2) $5\ \Omega$ (3) $7\ \Omega$ (4) $10\ \Omega$

45. An electric lamp is marked $60\ \text{W}$, $230\ \text{V}$. The cost of 1 kilowatt hour of power is Rs. 1.25. The cost of using this lamp for 8 hours is :-

- (1) Rs. 1.20
(2) Rs. 4.00
(3) Rs. 0.25
(4) Rs. 0.60

46. Equal volumes of 0.1M urea and 0.1M glucose solution are mixed. The mixture will have :-
 (1) Lower osmotic pressure
 (2) Higher osmotic pressure
 (3) Same osmotic pressure
 (4) Cannot be determined
47. The potential of the cell containing two hydrogen electrodes as represented below :
 $\text{Pt, H}_{2(g)}|\text{H}^+(10^{-6}\text{M})||\text{H}^+(10^{-4}\text{M})|\text{H}_{2(g)}, \text{Pt}$ at 298 K is :-
 (1) -0.118 Volt (2) -0.0591 Volt
 (3) 0.118 Volt (4) 0.0591 Volt
48. In one unit cell of NaCl if 4 Na^+ ions are present then Cl^- ions are :-
 (1) 2 (2) 8 (3) 4 (4) 6
49. n-hexane $\xrightarrow{\text{catalyst}}$ benzene
 catalyst is :-
 (1) AlCl_3 (2) $\text{SiO}_2-\text{Al}_2\text{O}_3$
 (3) $\text{Cr}_2\text{O}_3-\text{Al}_2\text{O}_3$ (4) MnO_2
50. $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{CH}_3 \xrightarrow[\text{hv}]{\text{Cl}_2} \text{Mono-chloro products} \xrightarrow{\text{Fractional Distillation}} \text{Fraction}$
 No. of products and no. of fractions are respectively
 (1) 6, 5 (2) 6, 4
 (3) 5, 4 (4) 6, 3
51. If α is degree of dissociation of Na_2SO_4 , the Van't Hoff factor (i) used for calculating the molecular mass is :-
 (1) $1 + \alpha$ (2) $1 - \alpha$
 (3) $1 + 2\alpha$ (4) $1 - 2\alpha$
52. For a zero order reaction, which of the following statement is false :-
 (1) The rate is independent of temperature of the reaction
 (2) The rate is independent of concentration of the reactants
 (3) The half life period depends on initial concentration of reactants
 (4) The rate constant has the unit $\text{mol litre}^{-1}\text{sec}^{-1}$
53. The rate constant of a reaction is expressed as
 $k = A e^{\frac{-40000}{T}}$, the energy of activation is :-
 (1) 40 kCal (2) 40000 kCal
 (3) 80 kCal (4) 80000 kCal
54. Which is correct order for the rate of sulphonation of benzene ?
 (1) $K_{\text{C}_6\text{H}_6} > K_{\text{C}_6\text{D}_6} > K_{\text{C}_6\text{T}_6}$
 (2) $K_{\text{C}_6\text{H}_6} < K_{\text{C}_6\text{D}_6} < K_{\text{C}_6\text{T}_6}$
 (3) $K_{\text{C}_6\text{H}_6} = K_{\text{C}_6\text{D}_6} = K_{\text{C}_6\text{T}_6}$
 (4) $K_{\text{C}_6\text{H}_6} > K_{\text{C}_6\text{D}_6} < K_{\text{C}_6\text{T}_6}$
55. Which of the following is not formed in given reaction

56. Under the same reactions conditions, initial concentration of $1.386 \text{ mol dm}^{-3}$ of a substance becomes half in 60 seconds and 30 seconds through first order and zero order kinetics, respectively.
 Ratio $\left(\frac{k_1}{k_0}\right)$ of the rate constant for first order (k_1) and zero order (k_0) of the reaction is :-
 (1) 1.0 mol dm^{-3}
 (2) $0.5 \text{ mol}^{-1}\text{dm}^3$
 (3) 1.5 mol dm^{-3}
 (4) $2.0 \text{ mol}^{-1}\text{dm}^3$

57. Freshly prepared precipitate sometimes gets converted to colloidal solution by :-

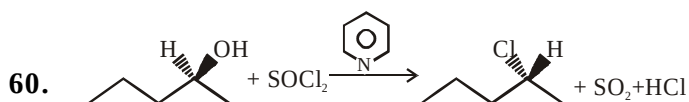
- (1) Coagulation (2) Electro osmosis
(3) Diffusion (4) Peptization

58. The radius of Ag^+ ion is 126 pm and I^- ion is 216 pm. The co-ordination number of Ag^+ ion is:-

- (1) 2 (2) 4 (3) 6 (4) 8

59. During mixed acid nitration of benzene Conc. HNO_3 behaves like :-

- (1) Acid (2) Base
(3) Catalyst (4) Electrophile



Proceed by the mechanism :

- (1) $\text{S}_{\text{N}}1$ (2) $\text{S}_{\text{N}}2$ (3) S_{E} (4) $\text{S}_{\text{N}}\text{i}$

61. 87g K_2SO_4 [At.wt of K = 39] is dissolve in 500g of H_2O then determine freezing point and boiling point of solution :-

Given $(K_f)_{\text{H}_2\text{O}} = 1.8 \text{ Kkgmol}^{-1}$

$(K_b)_{\text{H}_2\text{O}} = 0.5 \text{ Kkgmol}^{-1}$

- (1) 5.4°C , 101.5°C
(2) -5.4°C , 100.5°C
(3) -5.4°C , 101.5°C
(4) -1.8°C , 100.5°C

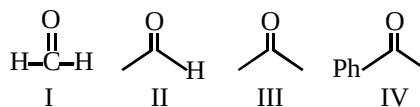
62. Physical adsorption of a gaseous species may change to chemical adsorption with :-

- (1) Increase in surface area of adsorbent
(2) Decrease in surface area of adsorbent
(3) Decrease in temperature
(4) Increase in temperature

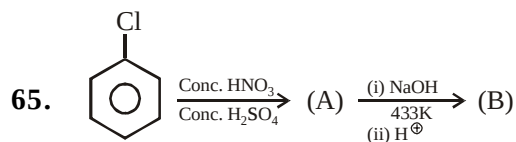
63. Fraction of total volume occupied by atoms in a SCC is:-

- (1) $\frac{\pi}{2}$ (2) $\frac{\sqrt{3}\pi}{8}$
(3) $\frac{\sqrt{2}\pi}{6}$ (4) $\frac{\pi}{6}$

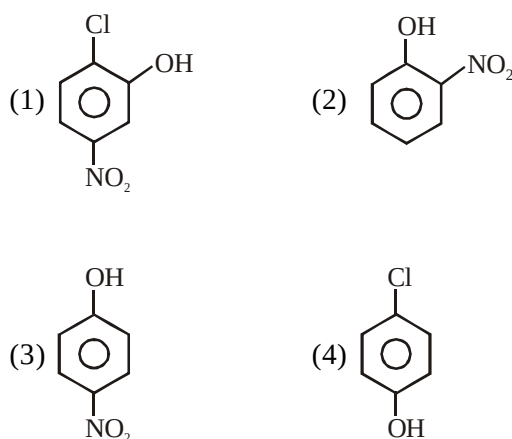
64. Arrange the following compounds in decreasing order of nucleophilic addition reaction :-



- (1) $\text{II} > \text{IV} > \text{III} > \text{I}$
(2) $\text{I} > \text{II} > \text{III} > \text{IV}$
(3) $\text{IV} > \text{III} > \text{II} > \text{I}$
(4) $\text{II} > \text{III} > \text{IV} > \text{I}$



B is a major product



66. Molarity of H_2SO_4 solution is 9M and density is 1.8 g/ml. Then molality of solution is :-

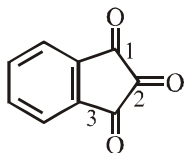
- (1) 19.8 (2) 9.8 (3) 6.8 (4) 4.5

67. Which of the following is an example of absorption and adsorption respectively ?

- (1) Water on silica gel & O_2 on charcoal
(2) Water vapours on CaCl_2 and air on charcoal
(3) H_2 on finely divided Ni and oxygen on metal surface
(4) Water on chalk surface and ink particles on chalk

68. How many unit cells are present in a cube shaped ideal crystal of NaCl of mass 2.00 g :-
 (1) 2.57×10^{21}
 (2) 1.28×10^{21}
 (3) 5.14×10^{21}
 (4) 1.71×10^{21}

69. Which carbonyl group is most reactive for nucleophilic addition reaction ?



- (1) 1
 (2) 2
 (3) 3
 (4) All have same reactivity
70. Which of the following do not give iodoform with ($I_2 + NaOH$) or NaOI ?
 (1) CH_3-CHO (2) CH_3-CH_2-CHO
 (3) CH_3-CH_2-OH (4) $CH_3-C(=O)-CH_3$

71. If standard reduction potentials of Ni^{+2}/Ni and Au^{+3}/Au half cells are -0.25 volt and 1.50 volt respectively, the E_{cell} of the cell, $Ni|Ni^{+2}(0.01M)||Au^{+3}(0.1M)|Au$ will be :-

- (1) Greater than E_{cell}°
 (2) Equal to E_{cell}°
 (3) Less than E_{cell}°
 (4) Unpredictable

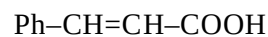
72. For the reaction, $2NO_2 \rightarrow N_2O_4 + O_2$, rate

expression is as follows $\frac{-d[NO_2]}{dt} = k[NO_2]^n$

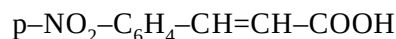
where $k = 3 \times 10^{-2} \text{ mol}^{-1} \text{ L sec}^{-1}$. If the rate of formation of oxygen is $1.5 \times 10^{-4} \text{ mol L}^{-1} \text{ sec}^{-1}$ then the molar concentration of NO_2 in mole L^{-1} is :-

- (1) $1.5 \times 10^{-4} \text{ M}$ (2) 0.01 M
 (3) 0.1 M (4) 0.316 M

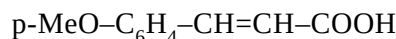
73. The order of decreasing reactivity in electrophilic addition of the following compounds is :-



(I)



(II)



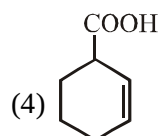
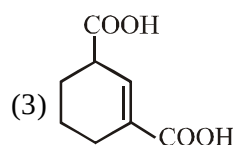
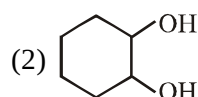
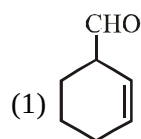
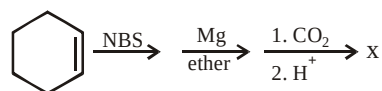
(III)

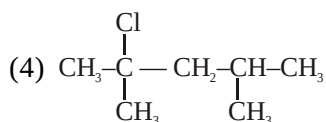
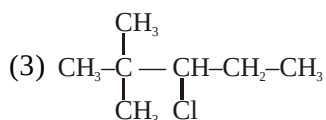
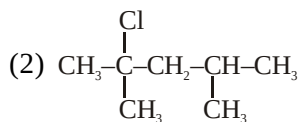
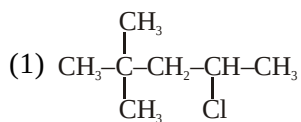
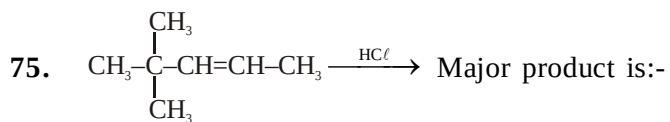


(IV)

- (1) $II > I > IV > III$
 (2) $III > IV > I > II$
 (3) $III > I > IV > II$
 (4) $II > IV > I > III$

74. In the given reaction (x) will be :-





76. 9.65 C of electric charge is passed through molten anhydrous MgCl_2 . The magnesium metal thus obtained is completely converted into Grignard's reagent (R-Mg-X). Calculate the number of moles of obtained Grignard's reagent :-

(1) 5×10^{-4}

(2) 1×10^{-4}

(3) 2×10^{-3}

(4) 5×10^{-5}

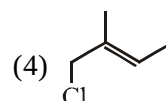
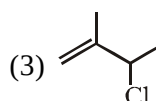
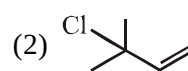
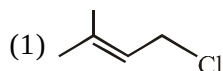
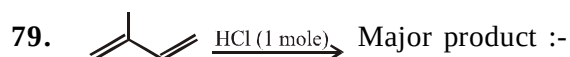
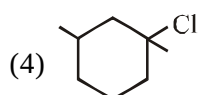
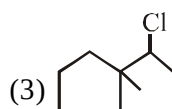
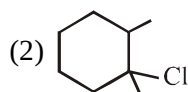
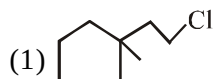
77. Which one is an example of gel ?

(1) Smoke

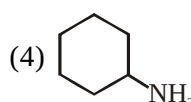
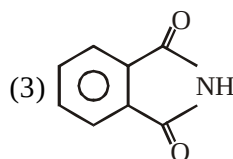
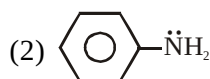
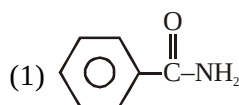
(2) Cheese

(3) Milk

(4) Fog



80. Which of the following is maximum basic :



81. Which of the following statements about solutions of electrolytes is not correct ?

- (1) Conductivity of solution depends upon size of ions
- (2) Conductivity depends upon viscosity of solution
- (3) Conductivity does not depend upon hydration of ions present in solution
- (4) Conductivity of solution increases with temperature

82. Which of the following is incorrect statement for adsorption :-

- (1) Physisorption is also known as vander wall's adsorption
- (2) Chemisorption is also known as activated adsorption
- (3) Chemisorption requires activation energy
- (4) Freundlich adsorption isotherm is $x/m = KP^{1/n}$ where $n \leq 1$

83. When isobutane is treated with Br_2 in sunlight, then major product is :-

- (1) 1° alkyl bromide
- (2) 2° alkyl bromide
- (3) 3° alkyl bromide
- (4) alkene

84. Which of the following structure has D-configuration ?

- | | |
|--|--|
| (1) $\begin{array}{c} \text{OH} \\ \\ \text{H}-\text{C}-\text{CHO} \\ \\ \text{CH}_2\text{OH} \end{array}$ | (2) $\begin{array}{c} \text{CH}_3 \\ \\ \text{OHC}-\text{C}-\text{OH} \\ \\ \text{H} \end{array}$ |
| (3) $\begin{array}{c} \text{CH}_2\text{OH} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{CHO} \end{array}$ | (4) $\begin{array}{c} \text{COOH} \\ \\ \text{CH}_3-\text{C}-\text{OH} \\ \\ \text{H} \end{array}$ |

85. $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{COOH} \xrightarrow{\Delta} (\text{A}) \xrightarrow[\text{HI}]{\text{Red P}} (\text{B})$ is

- | | |
|---|---|
| (1) $\text{CH}_3-\text{C}(=\text{O})-\text{OH}$ | (2) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$ |
| (3) $\text{CH}_3-\text{CH}_2-\text{CH}_3$ | (4) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3$ |

86. The resistance of 0.1 N solution of formic acid is 200 ohm and cell constant is 2.0 cm^{-1} . The equivalent conductivity (in $\text{S cm}^2\text{eq}^{-1}$) of 0.1 N formic acid is :-

- | | |
|---------|-------------------|
| (1) 100 | (2) 10 |
| (3) 1 | (4) None of these |

87. In Na_2O crystal, co-ordination number of Na^+ is :-

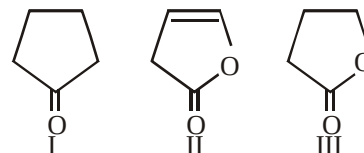
- | | | | |
|-------|-------|-------|-------|
| (1) 3 | (2) 4 | (3) 6 | (4) 8 |
|-------|-------|-------|-------|

88. $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{NOCl} \longrightarrow \text{P}$

Identify the product :-

- | | |
|-----|-----|
| (1) | (2) |
| (3) | (4) |

89.



Arrange the above compounds in the order of enol content

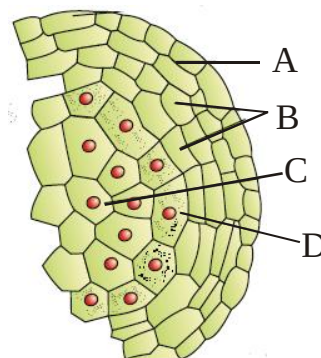
- | | |
|---|---|
| (1) $\text{I} > \text{II} > \text{III}$ | (2) $\text{III} > \text{II} > \text{I}$ |
| (3) $\text{II} > \text{I} > \text{III}$ | (4) $\text{II} > \text{III} > \text{I}$ |

90. + $\xrightarrow{\text{AlCl}_3} (\text{X})$

Major product is :-

- (1) $\text{Ph}-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$
- (2) $\text{Ph}-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}-\text{CH}_3$
- (3) $\text{Ph}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3$
- (4) $\text{Ph}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{CH}_3$

91. Cilia are observed in :-
 (1) Bronchiole
 (2) Stomach
 (3) Blood vessels
 (4) Alveoli
92. During swallowing which structure prevent entry of food into larynx ?
 (1) Glottis
 (2) Epiglottis
 (3) Thin hyaline cartilaginous flap
 (4) Thin fibromuscular flap
93. Wrong about haemodialysis is :-
 (1) Blood is drained from a convenient artery
 (2) Non-porous cellophane membrane is used
 (3) Cleared blood is pumped back to body through a vein after adding anti-heparin
 (4) Dialysing fluid have same composition as that of plasma except nitrogenous wastes.
94. Bioluminescence is characteristic of :-
 (1) *Taenia*, *Fasciola*
 (2) *Ctenoplana*, *Pleurobrachia*
 (3) *Obelia*, *Ctenoplana*
 (4) *Hydra*, *Ascaris*
95. Neurons are excitable cells because their membrane is in which state ?
 (1) Depolarised
 (2) Repolarised
 (3) Polarised
 (4) Hyperpolarised
96. Earthworm lack specialised breathing devices. Respiratory exchange occurs through ____ into their blood stream :-
 (1) Lungs (2) Moist body surface
 (3) Solenocytes (4) Flame cells
97. According to Darwin, the main mechanism of evolution is :-
 (1) Mutation (2) Saltation
 (3) Natural selection (4) Isolation
98. According to Darwin fitness means :-
 (1) Physical fitness
 (2) Reproductive fitness
 (3) Favourable mutations
 (4) Differential reproduction
99. Work of Cu-T
 (1) Prevent fertilization
 (2) Prevent implantation
 (3) Block the cervix
 (4) Prevent ovulation
100. Diagrammatic view of a microsporangium along with wall layer. Which part gives nutrition to developing pollen grain ?



- (1) D (2) B
 (3) A (4) C

101. Tissue having maximum intercellular spaces is :-
 (1) Dense connective tissue
 (2) Areolar connective tissue
 (3) Fluid connective tissue
 (4) Cartilage

102. Expiration takes place when :-

- (1) Intrapulmonary pressure is higher than the atmospheric pressure.
- (2) Diaphragm and EICM are contracted.
- (3) Intrapulmonary pressure is lower than the atmospheric pressure
- (4) Diaphragm and IICM are contracted.

103. Which mechanism helps to maintain a concentration gradient in the medullary interstitium?

- (1) Haldane mechanism
- (2) Heiring-Breuer mechanism
- (3) Counter-current mechanism
- (4) Cascade reaction

104. Sexual dimorphism is found in :-

- (1) *Hydra*
- (2) Earthworm
- (3) Leech
- (4) *Ascaris*

105. Which part of brain is associated with voluntary regulation of body muscles ?

- (1) Pons
- (2) Cerebrum
- (3) Mid brain
- (4) Cerebrum and pons

106. In pheretima a prominent dark band of glandular tissue (clitellum) is present in the segment numbers:-

- (1) 10, 11, 12
- (2) 12, 13, 14
- (3) 14, 15, 16
- (4) 15, 16, 17

107. Flipper of penguin and flipper of dolphin are example of :-

- (1) Homologous organ
- (2) Analogous organ
- (3) Vestigial organ
- (4) Atavismic organ

108. Find the incorrect match :-

	Column-I	Column-II	Column-III
(1)	Penis	Male external genitalia	Made up of three spongy cords
(2)	Seminal vesicles	One pair male sex accessory gland	Secretes seminal fluid
(3)	Bulbourethral gland	Also called bartholin gland	Secretes alkaline secretion
(4)	Leydig cells	Endocrine cells	Secretes testosterone

109. All of the following contraceptive method are hormonal except :-

- (1) LNG-20
- (2) Norplant
- (3) Multiload-375
- (4) Mala-D

110. Which set of plants produces two types of flowers?

- (1) Pea, *Oxalis*, *Commelina*
- (2) *Viola*, *Oxalis*, *Commelina*
- (3) *Parthenium*, Pea, *Commelina*
- (4) *Parthenium*, *Oxalis*, Brinjal

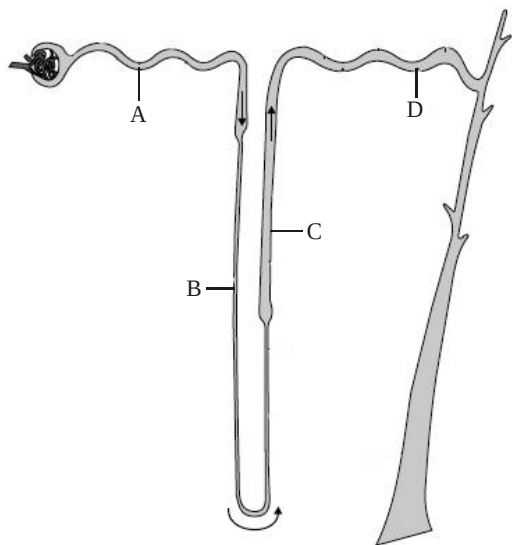
111. Tendons are example of :-

- (1) Dense regular connective tissue
- (2) Dense irregular connective tissue
- (3) Specialised connective tissue
- (4) Loose connective tissue

112. Additional volume of air, a person can expire by a forcible expiration is :-

- (1) IRV (2500-3000 ml)
- (2) ERV (1000-1100 ml)
- (3) ERV (1500-1600 ml)
- (4) EC (1500-1600 ml)

113. Part A, B, C, D of nephron are shown in the diagram. Select the option which is correct with it's function ?



- (1) A-Conditional reabsorption of salts
 (2) B-Impermeable to salt
 (3) C-Helps in reabsorption of nutrients
 (4) D-Reabsorb small amount of urea
114. Animals have 6 legs :-
 (1) Limulus, Cockroach
 (2) Scorpion, Spider
 (3) *Culex*, Julus
 (4) *Bombyx*, Laccifer
115. Hormone of endocrine gland not shows :-
 (1) Catalytic activity
 (2) Intercellular substance
 (3) Less secretion
 (4) Non-nutritive substance
116. In cockroach the 10th segment bears a pair of jointed filamentous structure called :-
 (1) Hypopharynx
 (2) Gonapophysis
 (3) Anal style
 (4) Anal cerci

117. Thorn of bougainvillea and tendril of cucurbita represents :-

- (1) Analogy
 (2) Homology
 (3) Atavism
 (4) Molecular homology

118. Find the incorrect match :-

	Column-I	Column-II
(1)	Uterus	Myometrium
(2)	Fallopian tube	Lined by ciliated epithelium
(3)	Clitoris	Made up of corpus spongiosum
(4)	Labia majora	Homologous to scrotum of male

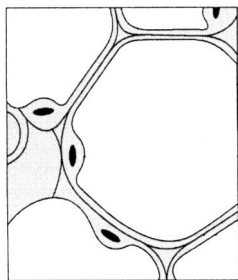
119. Read the following statements :-

- (a) Intentional termination of pregnancy before full term is called medical termination of pregnancy
 (b) Nearly 45 to 50 million MTPs are performed in a year all over the India
 (c) Government of India legalised MTP in 1977 with some strict conditions to avoid its misuse
- (1) Both a and c statement are incorrect.
 (2) Both b and c statement are correct.
 (3) Both a and b statement are incorrect.
 (4) Only a statement is correct.

120. White kernel in coconut represents the :-

- (1) Free nuclear endosperm
 (2) Cellular endosperm
 (3) Partially nuclear endosperm + Partially cellular endosperm
 (4) Embryo

121. Choose the incorrect for given diagram of a tissue:-



- (1) Type of loose connective tissue.
- (2) Located mainly beneath the skin.
- (3) Cells of this tissue store fats.
- (4) Fibres and fibroblast are compactly packed in this tissue

122. Asthma may be attributed to :-

- (1) Chronic disorder of alveolar wall
- (2) Difficulty in breathing associated with wheezing
- (3) Lung fibrosis
- (4) Decreased respiratory surface area

123. When any plane passes through central axis of body divides organisms into two identical halves is called :-

- (1) Animals with all types of symmetry
- (2) Asymmetrical
- (3) Bilateral symmetry
- (4) Radial symmetry

124. Urochordats have :-

- (1) Retrogressive metamorphosis
- (2) Notochord in larval stage
- (3) 1 and 2 both
- (4) Closed circulation with ventral heart

125. Cretinism is not characterised by :-

- (1) Low intelligence
- (2) Abnormal skin
- (3) Nodule formation
- (4) Mental retardation

126. Total number of bones in adult human :-

- (1) 306
- (2) 270
- (3) 260
- (4) 206

127. Which of the following is a correct statement ?

- (1) Evolution is a random process according to Darwin.
- (2) Lichens can be used as industrial pollution indicators.
- (3) Different aged rock sediments never contain fossil of different life forms.
- (4) Eyes of octopus and eye of mammals are example of homologous organ.

128. Which of the following statement is incorrect ?

- (1) The process of formation of a mature female gamete is called oogenesis.
- (2) Oogenesis is initiated just after the birth of girl child.
- (3) Tertiary follicle is characterised by a fluid filled cavity called antrum.
- (4) The secondary oocyte retain bulk of the nutrient rich cytoplasm of the primary oocyte

129. The medical termination of pregnancy (amendment) act 2017 was enacted by the government of India. According to this act a pregnancy may be terminated on more than 12 weeks, but fewer than 24 weeks of pregnancy on the opinion of :-

- (1) 4 registered medical practitioner
- (2) 1 registered medical practitioner
- (3) 6 registered medical practitioner
- (4) 2 registered medical practitioner

130. Which of the following is result of triple fusion ?

- (1) Zygote formation
- (2) Primary endosperm nucleus formation
- (3) Both zygote and endosperm formation
- (4) Neither zygote nor endosperm

131. Heart have :-

- (1) Only 3 types of tissues
- (2) Only 2 types of tissues
- (3) Only 1 tissue
- (4) All 4 types of tissues

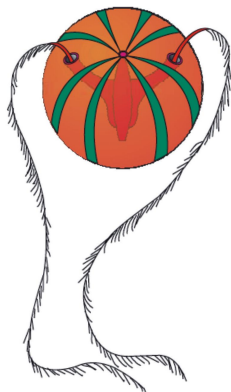
132. Which of the following blood cell can release variety of substances which involve in clotting of blood ?

- (1) Megakaryocyte (Platelet)
- (2) Macrophage
- (3) Kupffer cell
- (4) Mast cell

133. Find out the correct match :-

- (1) *Anopheles*, Octopus – Radula
- (2) *Plasmodium*, *Obelia* – Cellular grade organisation
- (3) *Paramecium*, *Plasmodium* – Organ grade organisation
- (4) Earthworm, Toad – Closed circulatory system

134. This animal is related with :-



- (1) Asexual reproduction
- (2) Bilateral symmetry
- (3) No digestion
- (4) Intra and extra cellular digestion

135. Hormone of gonads that is related to RBC formation:-

- (1) Cortisole
- (2) Androgen
- (3) ANF
- (4) Progesteron

136. Shoulder joint is :-

- (1) Ball and socket
- (2) Hinge
- (3) Pivot
- (4) Fibrous

137. Miller performed his experiment at which temperature:-

- (1) 400°C
- (2) 300°C
- (3) 200°C
- (4) 800°C

138. Menstrual cycle is controlled by :-

- (a) Estrogen and progesterone of ovary
- (b) FSH of anterior pituitary gland
- (c) LH of adenohypophysis
- (d) Oxytocin of neurohypophysis

Option :-

- (1) Only a and b are correct.
- (2) Only b and d are correct.
- (3) Only a and c are correct.
- (4) a, b and c are correct.

139. Find the correct match :-

	Column-I	Column-II	Column-III
(1)	Norplant	Hormonal method of contraception	Effective upto 10 years
(2)	Cu-7	Copper and hormone releasing IUD	Effective upto 1 year
(3)	Vaults	Physical barrier of contraception	Used by female
(4)	ICSI	Invivo fertilisation	A type of ART

140. Perisperm can be observed in seeds of :-

- (1) Black pepper
- (2) Castor
- (3) Pea
- (4) Both (1) and (2)

141. Select the incorrect statements :-

- (a) An adult human has 32 temporary teeth.
 - (b) Dental formula of human is $\frac{2123}{2123}$.
 - (c) Hard chewing surface of the teeth is made up of dentine
- (1) a, b
 - (2) a, c
 - (3) a, b & c
 - (4) b only

- 142.** Select the option which includes all the correct statements about lymph?
- Colourless fluid
 - Important carrier for nutrients, hormones
 - Glucose is absorbed through lymph in the lacteals
- Statement (a) and (c)
 - Statement (a) only
 - Statement (a) and (b)
 - Statement (b) only
- 143.** Animals which are bilateral symmetrical, triploblastic and pseudocoelomate with complete alimentary canal having well-developed muscular pharynx :-
- Mollusca
 - Platyhelminthes
 - Cnidaria
 - Nematoda
- 144.** Radula is found in :-
- Sea lily
 - Pila*
 - Sea urchin
 - Sea feather
- 145.** Insulin hormone of pancreas stimulates :-
- Gluconeogenesis, Protein synthesis
 - Protein synthesis, Lipolysis
 - Glycogenolysis, Lipogenesis
 - Glycogenesis, Protein synthesis
- 146.** Auto immune disorder affecting neuromuscular junction leading to fatigue, weakening and paralysis of skeletal muscle is :-
- Myasthenia gravis
 - Muscular dystrophy
 - Arthritis
 - Gout
- 147.** Miller used which compounds in his experiment:-
- CH_4 , NH_3 , H_2 , H_2O
 - CH_4 , O_2 , H_2O , H_2
 - CO_2 , O_2 , CH_4 , H_2
 - N_2 , CO_2 , H_2 , H_2O
- 148.** Which of the following event is correctly matched with time period in a normal menstrual cycle ?
- Ovulation $\rightarrow$ 17th day
 - Regeneration of endometrium $\rightarrow$ 5-10th days
 - Rise in progesterone level $\rightarrow$ 1-14th day
 - Endometrium degeneration $\rightarrow$ 10-15th day
- 149.** How many of the given statement(s) is/are false ?
- In most of angiosperms, pollen grains are shed at 2-celled stage.
 - Viability of pollen grains is lost within 30 minutes in rice.
 - Pollen tablets are used as food supplement
 - Egg apparatus possesses three synergids
- Four
 - One
 - Two
 - Three
- 150.** Both *Citrus* and mango are examples of :-
- One seeded fruits
 - Multiple seeded fruits
 - Nucellar polyembryony
 - Parthenocarpic fruits
- 151.** Which part of large intestine hosts some symbiotic micro-organism ?
- Ileum
 - Caecum
 - Oesophagus
 - Stomach
- 152.** When ventricular pressure falls during ventricular diastole, it causes :-
- Closure of semilunar valves
 - Closure of Atrio-ventricular valves
 - Opening of semilunar valves
 - Blood to flow through pulmonary and systemic aorta
- 153.** Protozoan animals are not related to :-
- Reproduction only by asexual means
 - Contractile vacuole
 - Food vacuole
 - Intracellular digestion

154. Which one is not a characteristic of mammals?

- (1) Presence of hairs
- (2) Presence of diaphragm
- (3) Presence of lungs
- (4) Presence of pneumatic bones

155. Miosis in eye stimulated by :-

- (1) Parasympathetic nervous system
- (2) Sympathetic nervous system
- (3) It does not respond to any type of nervous system
- (4) It is done by contraction of radial dilatory muscles

156. The process of evolution of different species in a given geographical area starting from a point and literally radiating to other area of geography (habitats) is called :-

- (1) Convergent evolution
- (2) Parallel evolution
- (3) Adaptive radiation
- (4) Mutation

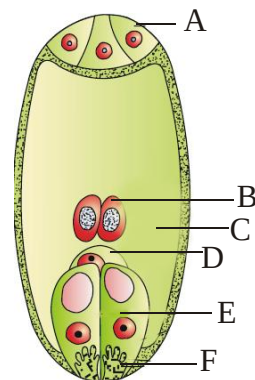
157. Factors which can affect Hardy-Weinberg equilibrium:-

- (1) Genetic drift
- (2) Mutation
- (3) Gene flow
- (4) All of the above

158. Which of the following is released by activated spermatozoa that acts on the ground substances of the corona radiata cells during fertilisation ?

- (1) Relaxin
- (2) Gonadotropin
- (3) Hyaluronidase
- (4) Trypsinogen

159. An embryo sac is shown in given diagram. Select the option with correct identification :-



	Filiform apparatus	Central cell	Antipodal cell
(1)	E	B	D
(2)	F	D	A
(3)	F	C	A
(4)	E	B	C

160. In polyembryony :-

- (1) Many seeds are present in a fruit
- (2) Many embryos are present in single seed
- (3) Fruits develop without fertilization
- (4) Seeds develop without fertilization

161. Which one of the following pairs of the type of cells and their secretion is correctly matched ?

- (1) β -cell of Islet of Langerhans – Glucagon
- (2) Chief cell of stomach – Proenzyme pepsinogen
- (3) Dust cell – Macrophage of liver
- (4) Pancreas – Gastrin

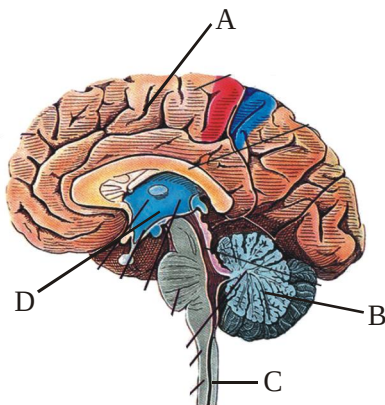
162. "Congestion of lungs" is associated with :-

- (1) Renal stone
- (2) Angina
- (3) Heart failure
- (4) Cholelithiasis

163. Contractile vacuoles for osmoregulation found in:-

- (1) *Plasmodium*, *Hydra*, *Amoeba*
- (2) *Amoeba*, *Paramecium*
- (3) *Paramecium*, *Obelia*, *Hydra*
- (4) *Balanoglossus*, *Saccoglossus*, Leech

164. Which part of brain is related with involuntary control of voluntary muscles ?



- (1) A
- (2) B
- (3) C
- (4) D

165. If light rays strike with iris then what will happen:-

- (1) It will enter in pupil
- (2) It will absorb by iris
- (3) It will cross iris and reaches upto fovea.
- (4) Light will form image on retina

166. Koala, sugar glider and tiger cat are examples of:-

- (1) Convergent evolution
- (2) Parallel evolution
- (3) Adaptive radiation
- (4) Analogy

167. A process in which heritable variation enabling better survival are enabled to reproduce and leave greater number of progeny is called :-

- (1) Natural selection
- (2) Genetic drift
- (3) Founder effect
- (4) Saltation

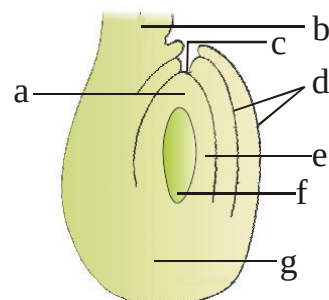
168. Attachment of the blastocyst in the wall of the uterus is known :-

- (1) Fertilisation
- (2) Placentation
- (3) Implantation
- (4) Syngamy

169. Usually pollen grains in angiosperms are shed at two celled stage but in few members, pollen grains are shed at three celled stage. These three cells are:-

- (1) 1 vegetative cell and 2 generative cells
- (2) 1 vegetative cell and 2 male gametes
- (3) 2 vegetative cells and 1 male gametes
- (4) 1 vegetative cell and 1 generative cell and 1 male gamete

170. Diagrammatic view of a typical ovule is given select the option with incorrect identification :-



- (1) d = Integuments
- (2) f = Embryosac
- e = Nucellus
- a = micropylar pole
- (3) b = Funicle
- (4) g = Embryosac
- g = Chalaza
- c = Hilum

171. Which of the following options best represents the enzyme composition of gastric juice ?

- (1) Amylase, Pepsin, Rennin
- (2) Peptidase, Lipase, Renin
- (3) Pepsin, Lipase, Rennin
- (4) Nuclease, Amylase, Pepsin

172. How much volume of CO_2 is removed by our lungs in 1 minute ?

- (1) 250 ml
- (2) 200 lit
- (3) 18 ml
- (4) 200 ml

173. Sypha is also known as :-

- (1) Sycon
- (2) *Hydra*
- (3) *Taenia*
- (4) *Scoliodon*

- 174.** Both neurofibrils and RER modified structure are found together in ____ and ____ of neuron.
(1) Cyton, Axon (2) Dendron, Axon
(3) Cyton, Dendron (4) Axon, Cyton
- 175.** Kinetic balance of body is regulated by :-
(1) Utriculus
(2) Semicircular canal
(3) Cochlea
(4) Eustachian tubule
- 176.** According to theory of special creation Earth is how much years old ?
(1) 1000
(2) 2000
(3) 3000
(4) 4000
- 177.** During evolution, Jawless fishes originated before how many years :-
(1) 350 million years ago
(2) 500 million years ago
(3) 400 million years ago
(4) 600 million years ago
- 178.** India is amongst the country in the world to initiate action plans and programmes at a national level to attain total reproductive health and child health care as a social goal. These programme are currently in operation under the popular name :-
(1) Family planing programme
(2) Reproductive aware programme
(3) Child aware programme
(4) Reproductive and child health care programme [RCH]
- 179.** Select the incorrect statement among followings:-
(1) Grasses are commonly pollinated by wind
(2) Bees are the most common biotic pollinating agents
(3) Pollen grains are sticky in wind pollinated flowers
(4) Self incompatibility is an out-breeding advice
- 180.** Select the wrongly matched pair :-
(1) Ovary → Fruit
(2) Integuments → Seedcoat
(3) Ovule → Seed
(4) Nucellus → Endosperm



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